SIGNAL GENERATION AND BASIC SIGNAL OPERATIONS

Experiment 1.1

Generation of Continuous Time Signals

Aim

To verify the generation of a sinusoidal signal with a specific frequency, amplitude, and phase shift using MATLAB. The Aim is to ensure that the signal generator can accurately produce a sinusoidal waveform with the desired parameters.

Apparatus Required

PC with MATLAB.

Program:

```
% Generation of Continuous Time Signals

clc;

clear;

close all;

% Time axis

t = -5:0.01:5;

% Continuous-time sine signal

x = sin(2*pi*t);

% Plotting

figure;

plot(t, x, 'LineWidth', 2);

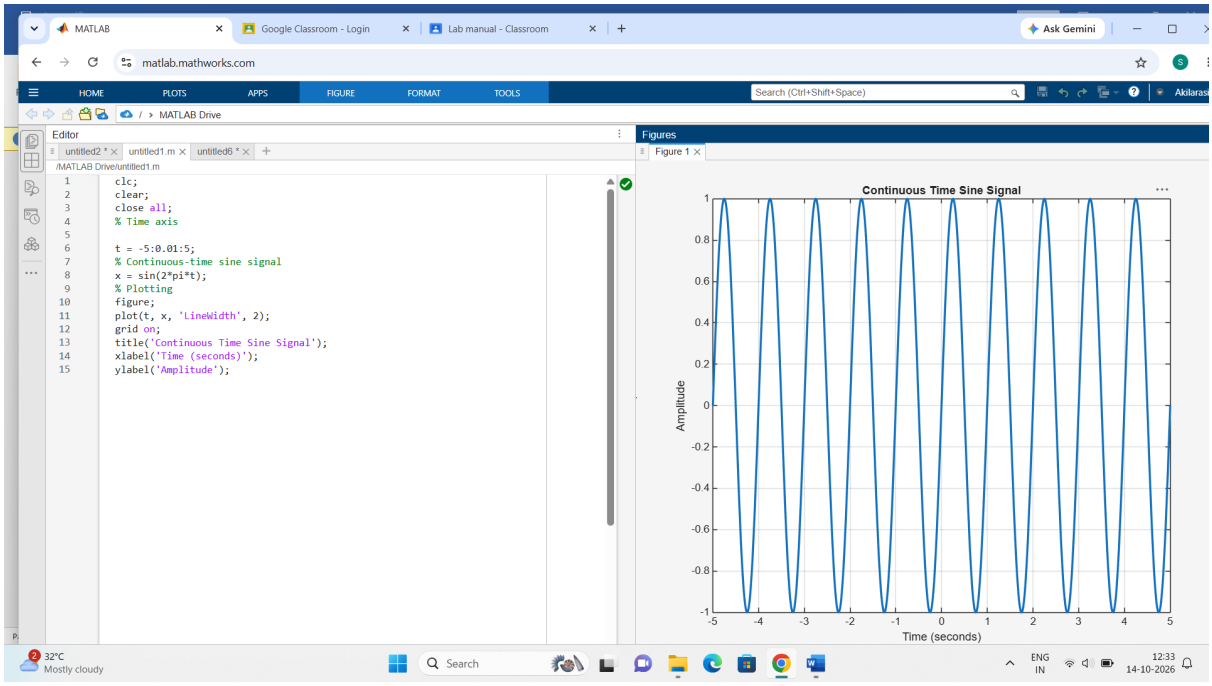
grid on;

title('Continuous Time Sine Signal');

xlabel('Time (seconds)');

ylabel('Amplitude');
```

Output Waveform:



Result:

This test case validates that the signal generator can produce a sinusoidal waveform with the specified frequency, amplitude, and phase shift.